

HMD Type and Spatial Ability: Effects on the Experiences and Learning of Students in Immersive Virtual Field Trips

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ABSTRACT

We report on the results of a study in the context of place-based immersive VR (iVR) geoscience education that compares the experiences and learning of 45 students after going through an immersive virtual field trip, using either a lower-sensing but scalable Oculus Quest or a higher-sensing but tethered HTC Vive Pro. Our results suggest that with content design considerations, standalone HMDs can be a viable replacement for high-end ones in large-scale educational studies. Furthermore, our results also suggest that the spatial ability of students can be a determining factor for their experiences and learning.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality; Applied computing—Education—Interactive learning environments

1 INTRODUCTION

With the emergence of cost-effective immersive head-mounted displays (HMDs), we have observed a radical shift from conventional virtual experiences such as desktop applications (sometimes also referred to as VR in the literature) and virtual reality Cave Automatic Virtual Environments (CAVE) systems, to fully immersive virtual reality (iVR) experiences facilitated through the power of HMDs. As argued by Dede [1], HMDs offer total sensory immersion since users can touch virtual objects using interactive cues that are natural to them, and the virtual experience reacts to the motions of the user (i.e., head movements and physical translations in the space). Given the range of HMDs offered in the market these days, and considering their associated costs, requirements (i.e., tethered to a VR-ready computer, or standalone), and capabilities (e.g., display resolution, hand tracking, degrees of freedom (DOF), field of view (FOV), etc.) [5], one of the pertinent questions in the immersive learning research domain is to what extent the associated attributes of different HMDs affect their applicability in education? And whether the superiority of high-end HMDs in terms of tracking and sensing justify their shortcomings in scalability and mass dissemination in educational and research settings [5]. In light of this, we have explored the applicability of two different VR HMDs (Oculus Quest and HTC Vive Pro) in an educational immersive virtual field trip (iVFT) that focuses on geoscience education.

2 EXPERIMENT

The iVFT used in this research takes students to the Reedsville and Bald Eagle Formations as part of an introductory geoscience course at Penn State. The iVFT is based on 360-degree high-resolution images, diagrams, and a 3D model of an outcrop built using structure from motion. The entire iVFT is narrated with audio clips that

explain features of the outcrop and the tasks learners need to perform (see [4] for more details).

A between-subject design was used to explore the effect of HMD (Oculus Quest vs. HTC Vive Pro) on the experiences and learning of participants in the iVFT. A total of 45 students, of which 30 (19 males, 10 females, and 1 other, with an average age of 19.7) went through the iVFT using the Oculus Quest and 15 (8 males and 7 females, with an average age of 19.2) went through the iVFT using the HTC Vive Pro, participated in this study. All students were recruited from a geoscience class at Penn State and were randomly assigned to either condition. The iVFT experience was embedded in this course as a class assignment and students were awarded course credit for their participation.

The experiences and learning of the participants were assessed using self-reported questionnaires. As part of the demographic information, participants were asked to report on their sense of direction (SOD; i.e., environmental spatial ability) using the Santa Barbara Sense of Direction Scale (SBSOD) [2].

After going through the iVFT, the experiences of the participants were measured in light of simulator sickness [3], and spatial situation model (SSM) and self-location (as two levels of developing spatial presence through experiences in the virtual field site [9]). The learning experience and perceived learning outcome of the participants were measured using the questionnaires from Lee et al. [6], in light of representational fidelity, immediacy of control, perceived usefulness, perceived ease of use, field trip enjoyment/motivation, cognitive benefits, control and active learning, reflective thinking, perceived learning effectiveness, and satisfaction. All metrics were measured on a scale of 1 to 5, and individual items were averaged and collapsed into corresponding final metric scores.

Welch's two-sample t-tests were used to explore the effects of HMDs on the experience and learning of participants. The two-sample Mann-Whitney-Wilcoxon test was used if statistical assumptions were violated. Our results show that differences in the experiences of users in the Oculus Quest and HTC Vive Pro conditions were significant with respect to simulator sickness with Quest participants ($M = 1.95$, $SD = .73$) experiencing significantly more simulator sickness than the Vive Pro participants ($M = 1.58$, $SD = .42$), $t(24.57) = 39.84$, $p = .03$, Cohen's $d = .58$. It is important, however, to note that the average scores for simulator sickness in both conditions were quite low. Furthermore, with respect to satisfaction, Quest participants ($M = 3.8$, $SD = .84$) reported significantly higher satisfaction than Vive Pro participants ($M = 3.21$, $SD = .87$), $t(24.57) = 2.10$, $p = .046$, Cohen's $d = .69$. The differences between the two conditions for all the other measured metrics were statistically insignificant.

Given the larger sample size in the Quest condition, we explored the effect of individuals' spatial ability using only the results from the Quest participants. Merging the responses from both conditions was avoided due to the significant differences reported between them in the t-test analyses.

As part of the demographic questionnaire, we measured the 30 Quest participants' sense of direction (SOD). We performed a median split to convert the numeric scores into two categories: low-

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SOD and high-SOD participants. The experience and learning metrics of participants in the Quest condition were then compared based on their SOD category. Our results showed that participants' SOD had a significant effect on several of the experience and learning metrics. More precisely, high-SOD participants ($M = 4.05$, $SD = .63$) reported significantly higher spatial situation model scores than low-SOD students ($M = 3.16$, $SD = .77$), $t(25.23) = 3.39$, $p = .002$, Cohen's $d = 1.27$. High-SOD participants ($M = 3.97$, $SD = .68$) also reported significantly higher self-location scores than low-SOD students ($M = 3.26$, $SD = .70$), $t(26.72) = 2.77$, $p = .01$, Cohen's $d = 1.03$.

With respect to the learning metrics, high-SOD participants ($M = 4.07$, $SD = .56$) reported significantly higher perceived usefulness than low-SOD students ($M = 3.25$, $SD = .85$), $t(22.35) = 3.03$, $p = .006$, Cohen's $d = 1.14$. High-SOD participants ($M = 4.07$, $SD = .46$) also had significantly higher reflective thinking scores than low-SOD students ($M = 3.04$, $SD = .91$), $t(18.9) = 3.82$, $p = .001$, Cohen's $d = 1.45$. Lastly, high-SOD participants ($M = 3.91$, $SD = .60$) reported significantly higher perceived learning effectiveness than low-SOD students ($M = 3.21$, $SD = .98$), $t(21.16) = 2.29$, $p = .03$, Cohen's $d = .86$.

3 DISCUSSION AND CONCLUSION

Our results indicate that although participants in the Quest condition reported higher levels of simulator sickness, the average score for this metric was quite low in both conditions (1.95 for Quest and 1.58 for Vive Pro). Furthermore, participants in the Quest condition reported significantly higher levels of satisfaction compared to those in the Vive Pro condition. The observed differences in simulator sickness scores could be due to the inferiority in the display resolution, frame rate, and the FOV of the Quest compared to Vive Pro. Such inferiority might manifest itself in the perception of the 360-degree images and the 3D model of the outcrop used for the hands-on activities. Further, in our study the Oculus Quest had a smaller play area than the Vive pro, which could have negatively affected the simulator sickness scores of Quest participants. Considering that the other experience and learning metrics of participants were not significantly different between the conditions, our results suggest that the Oculus Quest can be considered as a viable and cost-effective HMD for learning via iVFTs. Our findings have important implications in the academic research domain and for classroom education with iVR technology. The logistics of experiencing an iVFT using the Vive Pro are significantly more complicated for both the experimenters as well as the students [5]. Evidently, an HMD that requires a VR-ready computer can not be easily transported into lab sessions or classrooms, which leads to asking students to interrupt their schedule and go to a specified location to do their assignment. Although a minor inconvenience on the students' part, this requirement could introduce a negative effect on the satisfaction level of the students with the iVFT as a whole.

There is a division of opinion in the literature on whether the HMD type actually affects the experiences of learners. Our results indicate that such differences do exist. Although inline with the results reported by [8] (comparing Oculus Rift with HTC Vive), we have compared a standalone HMD versus a tethered one. Therefore, our results not only pertain to the quality comparison of HMDs, but are also relevant and important in understanding the potential of standalone HMDs in large-scale classroom applications. Our work partially confirms the findings reported by [7], in that standalone devices are as good as more advanced HMDs in educational environments. We complement these results (in the context of geosciences VFTs) with the findings reported above highlighting that standalone and scalable HMDs seem to cause higher levels of satisfaction in users than high-end tethered ones, probably mostly due to practical advantages.

Our study has highlighted the importance of scalability in large-

scale classroom user studies [5]. Data collection with high-end HMDs is associated with higher costs, not only in terms of equipment but also time, since evaluations need to be performed with one participant at a time. We attempted to streamline this process with the help of the VR center at the library of our institution. However, this process introduced a new series of challenges such as the lack of a constantly present experimenter for data collection to effectively moderate the evaluation sessions. These logistical challenges further demonstrate the advantages of scalable HMDs in evaluations and integration of iVFTs within classrooms.

Our findings also indicate that individual differences between learners, their sense of direction, can be a significant determinant of their experiences and learning in an iVFT. It was shown that high-SOD participants had a better experience in light of the spatial situation model, self-location, perceived usefulness, reflective thinking, and perceived learning effectiveness compared to low-SOD participants. It is important to note that we have not measured the learning outcome/performance of participants in the context of this study and, therefore, cannot draw any conclusions on whether low-SOD participants have learned the same from the iVFT compared to the high-SOD ones. Notwithstanding, our findings indicate that environmental spatial ability is an important learner characteristic that should be considered in the design of iVFTs. This opens the door for numerous interesting research questions related to design decisions for immersive experiences with the objective of catering to both high and low spatial ability learners.

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